

ROBUST CONFIDENCE INTERVALS FOR SCALE AND ITS APPLICATION TO THE RAYLEIGH DISTRIBUTION

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Summary: Some general results concerning the determination of robust interval estimators for the parameter, θ , of a scale family of distributions, $F(x/\theta)$, are developed. Specifics are discussed in the case when sampling takes place from the Rayleigh family and when the sample could be contaminated by inlying or outlying observations. The choice of point estimator on which the confidence interval is based directly reflects on the robustness qualities of the interval estimator. Several choices are considered, i.e. the MAD, Hampel's M-estimator and the trimmed maximum likelihood estimator. The quality of interval estimators is evaluated on two criteria simultaneously, viz. coverage probability and expected length. If no contamination or inlier contamination is present, the use of the customary maximum likelihood estimator is recommended for computing the confidence interval. The estimator of choice when up to 10% outlier contamination occurs is Hampel's M-estimator.

1. INTRODUCTION

The confidence interval (CI) is one of the most important tools of modern data analysis, both from the classical and robustness points of view. The

earlier work on robust estimation (starting with Huber, 1964, section 10, and ranging through Hampel, Ronchetti, Rousseeuw and Stahel, 1986) dealt almost exclusively with point estimation of location parameters, the estimation of scale playing a secondary role (and for obvious reasons). The main advantage of robust methodology is that it provides protection against unknown deviations from the assumed underlying model – typically against outliers. Interval estimation of location is critically dependent on a quality robust estimator of scale – the former is sunk if a poor scale estimator is used, and this aspect has received attention in the literature, see for example Iglewicz (1983). In other cases, scale is the primary parameter of interest, and we will focus here on that situation. Point estimation of scale (Kidd and Laubscher, 1993) is further developed to a corresponding theory for interval estimation. Due to the parametric nature of infinitesimal robustness models, some of the theoretical results are general but much of its practical implementation depends on the details of the underlying model. The family selected as a vehicle for this presentation is the Rayleigh family, as in our paper on point estimation for scale; cf. Kidd and Laubscher (1993), where the practical relevance of that choice was motivated. An important issue in this respect is that the Rayleigh is an interesting representative of the families of *non-symmetric* scale distributions which are of special interest in the context of robust estimation of scale.

In sections 2 and 3 we propose and develop a criterion for the construction of CIs based upon a robust estimator of scale. Several well-known robust estimators are briefly reviewed in section 4 and applied in the construction of robust CIs in section 5. Section 6 discusses ways of evaluating the performance of selected estimators and section 7 applies this to a simulation study. The evaluation is done for the pure underlying Rayleigh family as well as for mixtures leading to outlier as well as inlier contamination. Recommendations are made as to the estimator of choice in different circumstances.

2. GENERAL RESULTS

In this section we study the construction of *robust* confidence intervals for the scale parameter θ of a random sample $X_1, \dots, X_n$ from a continuous underlying scale family of distributions described by $F_\theta(x) = F(x/\theta)$ with density function $\theta^{-1}f(x/\theta)$, based on some (robust) estimator T_n . The estimators discussed in this paper all have the property of *scale equivariance* which may be written symbolically as $T_n(aX) = |a| T_n(X)$ for any scale constant a .

To construct confidence intervals, we firstly have to know the distribution of T_n , say H_θ , which is naturally dependent on the underlying distribution F_θ . Let H be the distribution of T_n when the standard member $F \equiv F_{\theta=1}(x)$ is the underlying distribution. To be able to construct a confidence interval for θ we have to find the relationship between H_θ and H . It is not difficult to establish the result that H_θ itself is a member of the scale family, namely one that satisfies the relationship $H_\theta(t) = H(t/\theta)$. The proof of this is developed by noting that the mechanism that generates a random sample $X_1, \dots, X_n$ from F_θ corresponds to a set of uniform random numbers, say $U_1, \dots, U_n$, via the probability integral transformation. These, in turn (by the inverse probability integral transformation) will correspond to a random "sample" $X_1/\theta, \dots, X_n/\theta$ from the standard member F . Let $T_n = T_n(X_1, \dots, X_n)$, and $S_n = T_n(X_1/\theta, \dots, X_n/\theta)$. From the scale equivariance property it follows that $S_n = T_n/\theta$. Then for any realization $T_n = t$,

$$H_\theta(t) \equiv P(T_n \leq t) = P(\theta S_n \leq t) = P(S_n \leq t/\theta) = H(t/\theta)$$

and

(2.1)

$$H_\theta^{-1}(u) = \theta H^{-1}(u).$$

This relationship (with F assumed known) enables the construction of a $1 - \alpha$ confidence interval for θ :

(2)

$$\begin{aligned} 1 - \alpha &= P(H_\theta^{-1}(\alpha_1) \leq T_n \leq H_\theta^{-1}(\alpha_2)) \\ &= P(\theta \cdot H^{-1}(\alpha_1) \leq T_n \leq \theta \cdot H^{-1}(\alpha_2)) \\ &= P(T_n/H^{-1}(\alpha_2) \leq \theta \leq T_n/H^{-1}(\alpha_1)) \end{aligned}$$

where α_1 and α_2 are chosen such that $\alpha_2 - \alpha_1 = 1 - \alpha$. Thus

$$(2.3) \quad (T_n/H^{-1}(\alpha_2); T_n/H^{-1}(\alpha_1))$$

forms the $(1 - \alpha)$ CI for θ . Note that the robustness properties of T_n are directly inherited by its corresponding confidence interval. We now set about finding suitable values for α_1 , α_2 and the quantiles of H .

3. THE QUANTILES OF H

The confidence interval for θ is determined by solving (2.2) for suitably selected values of α_1 and α_2 . The usual choices are $\alpha_1 = \alpha/2$ and $\alpha_2 = 1 - \alpha/2$. This works well if the distribution of the estimator is symmetric. For scale estimators this is not always the case. Another feasible way of choosing α_1 and α_2 , introduced here, is to select their values in such a way that θ (being estimated by T_n) lies midway between the two quantiles $H_\theta^{-1}(\alpha_1)$ and $H_\theta^{-1}(\alpha_2)$ of T_n , see figure 3.1. This means that

$$(3.1) \quad H_\theta^{-1}(\alpha_2) - \theta = \theta - H_\theta^{-1}(\alpha_1)$$

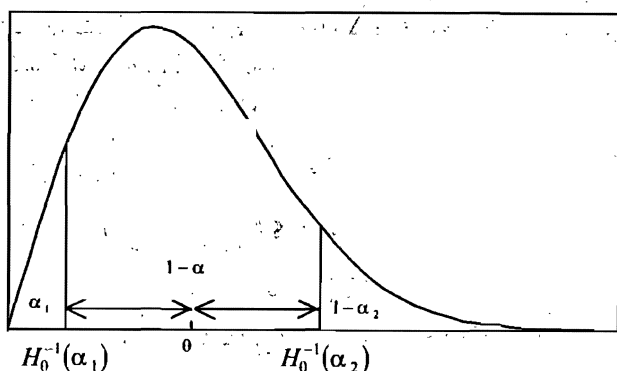


Figure 3.1. The density of T_n . The tail probabilities α_1 and $1 - \alpha_2$ are chosen in such a way that θ lies in the centre of the interval spanned by the quantiles $H_\theta^{-1}(\alpha_1)$ and $H_\theta^{-1}(\alpha_2)$.

or in terms of the standard member: $H^{-1}(\alpha_2) - 1 = 1 - H^{-1}(\alpha_1)$. This method is more general than the first, and if the distribution turns out to be symmetric, then the two methods are the same.

The next step in solving (2.2) is to find H , the distribution of T_n under F . A practical difficulty now arises in that H , being the standard member of the distribution of T_n , will be dependent on the sample size n . This creates a problem in handling the quantiles for different sample sizes. The way in which this will be dealt with is to fit a smooth curve linking the quantiles of H and sample size n following a simulation for the distribution H ; for an example, see figure 5.1. This relationship can be motivated analytically as follows:

Consider the case $\theta = 1$. For most of the estimators, T_n , we have asymptotic normality (cf. Hampel, Rousseeuw, Ronchetti and Stahel, 1986), i.e.

$$(3.2) \quad \mathcal{L}_F(\sqrt{n}(T_n - 1)) \xrightarrow{n \rightarrow \infty} N(0; V(T; F)),$$

where $V(T, F)$ is the asymptotic variance of the functional T with F as underlying distribution. Here T denotes the functional corresponding to T_n i.e. $T_n = T(F_n)$ either exactly or approximately with F_n the empirical distribution function of the data. Let us denote the α quantile of H by $x_n(\alpha) := H^{-1}(\alpha)$, in which the notation exposes the dependence of the distribution of H on the sample size. Hence, for large n it follows from (3.2) that

$$(3.3) \quad \begin{aligned} \alpha &= H(x_n(\alpha)) \\ &= P(T_n \leq x_n(\alpha)) \\ &= P(\sqrt{n}(T_n - 1) \leq \sqrt{n}(x_n(\alpha) - 1)) \\ &\approx \Phi \left(\frac{\sqrt{n}(x_n(\alpha) - 1)}{\sqrt{V(T; F)}} \right). \end{aligned}$$

Hence

$$(3.4) \quad x_n(\alpha) = H^{-1}(\alpha) \approx 1 + \Phi^{-1}(\alpha) \sqrt{V(T; F)/n}.$$

This means that for large n , $x_n(\alpha)$ is approximately linear in $1/\sqrt{n}$.

and the form of this expression will be used to specify the non-linear function (3.5) below, to determine the quantiles for given sample size.

As mentioned earlier, simulation studies were used to approximate H because it is unknown for most of the estimators and difficult to determine analytically. For sample sizes varying from 10 to 100 in steps of 10, $N = 100\,000$ samples were pseudo-randomly generated from F . The value of T_n was calculated for each of the samples and H_N , the empirical distribution function of H based on the T_n 's, used to approximate H . For fixed choices of α , curves are fitted through the empirically determined points $H_N^{-1}(\alpha_1)$; $H_N^{-1}(\alpha_2)$ versus n . The functions used for the curve fitting were selected to be a slight generalization of the form suggested by (3.4) as

$$(3.5) \quad H_N^{-1}(\alpha_1) = 1 - c_0/n^{c_1} \text{ and } H_N^{-1}(\alpha_2) = 1 + c_0/n^{c_1},$$

for some unknown constants c_0 and c_1 . The change in the sign of the second part of equation (3.5) (from minus to plus) stems from the way α_1 and α_2 were chosen, viz. such that the probability is highest for θ to lie in the centre of the interval spanned by $H^{-1}(\alpha_1)$ and $H^{-1}(\alpha_2)$.

Having determined values for $H_N^{-1}(\alpha_1)$ and $H_N^{-1}(\alpha_2)$ we can now use this, in (2.2) to obtain (approximate) CIs for θ . In section 5 we shall apply this approximation to evaluate the confidence intervals.

4. A SELECTION OF SCALE ESTIMATORS

The Rayleigh family, specified below, is an asymmetric member of the general scale family of distributions. It is defined on the positive real line, which makes it special from the point of view of M-estimation which has by and large been applied to families of underlying symmetric distributions. It underlies an interesting class of practical situations, alluded to in the introduction, where the robust estimation of its parameter is relevant (cf. Kidd, 1991). The density function of the Rayleigh family is

$$(4.1) \quad r_\theta(x) = 2x\theta^{-2} \exp(-(x/\theta)^2), \quad \text{with } x, \theta \in \mathbb{R}^+.$$

The distribution and density functions of the standard member of the Rayleigh are

$$(4.2) \quad R(x) = 1 - \exp(-x^2) \quad \text{and} \quad r(x) = 2x \exp(-x^2)$$

respectively. The Fisher score functions for the Rayleigh and its standard member are,

$$(4.3) \quad s_\theta(x) = \frac{\partial}{\partial \theta} \ln(r_\theta(x)) = (2/\theta)[(x/\theta)^2 - 1], \quad \text{and} \quad s(x) = 2(x^2 - 1).$$

In what follows we discuss and apply several competing point estimators for θ with a view to constructing CIs. As can be seen in (2.3) the CI is directly related to the choice of point estimator T_n . The estimators included in this study are all selected on the basis of favourable robustness properties. Their influence functions (for the Rayleigh family) were studied by Kidd (1991). They all have bounded gross error sensitivity and their finite sample breakdown points are above 0.4. The maximum likelihood estimator (MLE) does not have the robustness properties discussed above, but was included for comparative purposes. We identify the estimators studied here for brevity by appending the character "R" (abbreviating *Rayleigh*) to the end of the name of the estimator.

4.1 Hampel M-estimator (*HampelR*)

This M-estimator was introduced by Kidd and Laubscher (1993) and is derived through the application of Hampel's Lemma 5 (cf. Hampel, 1968). The ψ -function for the M-estimator is

$$(4.4) \quad \psi(x; \theta) = [2(x/\theta)^2 - a(b) - 2]_{-b}^{+b}$$

for a chosen trimming constant $b > 0$. The function

$$(4.5) \quad a(b) = \left\{ a \left| \int_0^a [2x^2 - a - 2]_{-b}^{+b} r(x) dx = 0 \right. \right\}$$

ensures its Fisher consistency. The trimming constant is selected as $b = 1.0$ (and is optimal from the point of view of minimizing mean squared error). This choice is based on results from Kidd and Laubscher (1993). Hence, as is

customary with M-estimators, the *HampelR* for a random sample $\{X_i\}$ is computed as the solution of θ from the equation

$$(4.6) \quad \sum_{i=1}^n [2(X_i/\theta)^2 - a - 2]_{-1}^{+1} = 0,$$

where the Fisher consistency constant is obtained as the solution of a from the equation

$$(4.7) \quad \int_0^{\infty} x e^{-x^2} [2x^2 - a - 2]_{-1}^{+1} dx = 0$$

giving $\hat{a} = -0.5311$.

4.2 MAD (*MadR*)

This estimator was studied in a general context by various authors (see e.g. Hampel et al., 1986). It is an M-estimator, with estimating equation $\psi_{\text{MAD}}(x) = \text{sign}(x - a)$ where a is a constant ensuring its Fisher consistency. When applied to the Rayleigh family the *MadR* reduces to $T_n = a^{-1} \text{Med}\{X_i\}$ with $a = \sqrt{\pi/2} = 0.8326$. Hence, for a random sample from an underlying Rayleigh distribution we may compute

$$(4.8) \quad \text{MadR} = \text{Med}\{X_i\}/0.8326.$$

This estimator differs from the better-known class of MAD estimators (where the median of the absolute deviations from the median is calculated) in that it actually turns out to be nothing more than a scaled median. We have decided to continue calling it a MAD estimator since it is derived from the same ψ -function as the classical MAD and does therefore belong to the same class.

4.3 Cauchy estimator (*CauchyR*)

Using the Fisher score function (a bounded function) of the Cauchy distribution, we construct an M-estimator with estimating equation:

$$(4.9) \quad \psi_c = 2x^2/(1 + x^2) - a$$

where $a = 0.8073$ is the Fisher consistency constant. This estimator was introduced and studied by Kidd (1991).

4.4 The interpoint distance estimator (Q_nR)

This estimator (Q_n , introduced by Rousseeuw and Croux, 1993) is defined as the k -th order statistic of pair-wise differences in the sample:

$$(4.10) \quad T_n(X_1, \dots, X_n) := d \cdot \left\{ |X_i - X_j|; i < j \right\}_{(k)}$$

where d is the Fisher consistency constant, and $k = \left\lfloor \frac{h}{2} \right\rfloor \approx \left\lfloor \frac{n}{2} \right\rfloor / 4$ where $h = \lceil n/2 \rceil + 1$ is roughly half the number of observations. A fast algorithm for calculating (4.10) proposed by Rousseeuw and Croux (1993) was used and the value of d for the Rayleigh distribution was determined numerically to be 4.91.

4.5 Trimmed maximum likelihood ($TrMLE$)

Since the MLE for the Rayleigh parameter is $\hat{\theta} = \sqrt{\frac{1}{n} \sum_{i=1}^n X_i^2}$, the α -trimmed MLE (see Kidd, 1991) is defined as:

$$(4.11) \quad T_n := \frac{1}{a} \sqrt{\frac{1}{n-2\lceil \alpha n \rceil} \sum_{i=\lceil \alpha n \rceil+1}^{n-\lceil \alpha n \rceil} X_i^2}$$

with $\alpha < 0.5$ and Fisher consistency constant a determined as:

$$(4.12) \quad a = \sqrt{1 + \frac{\alpha \ln(\alpha) - (1-\alpha) \ln(1-\alpha)}{1 - 2\alpha}}$$

From Kidd (1991) an optimum selection for α based on standardized mean squared error is $\alpha = 0.4$, which leads to $a = 0.8966$ as Fisher consistency constant.

We also included the MLE in our simulation study for comparison purposes and denoted it by $MleR$.

5. CONSTRUCTING THE CONFIDENCE INTERVALS

In this section we construct confidence intervals for the scale parameter of the Rayleigh distribution using the scale estimators discussed in section 4.

These confidence intervals are then compared in terms of real coverage probability and interval length under various types of contamination.

Figure 5.1 depicts the relationship between the sample size and the quantiles of the distribution of the *HampelR* estimator for $\theta = 1$.

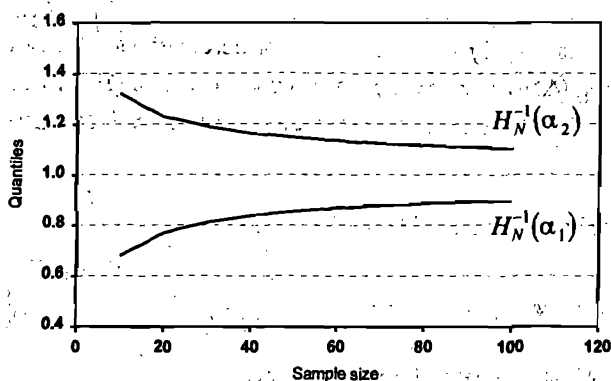


Figure 5.1. Relationship between sample size and the quantiles of the distribution of the *HampelR* estimator for $\theta = 1$.

Table 5.1 gives the least squares fitted values for c_0 and c_1 from (3.5) for the estimators and for coverage probabilities ranging from 0.90 to 0.99.

From table 5.1 one can see that all the values of c_1 are relatively close to 0.5 as may be expected from (3.4). The values of c_0 obtained after fitting (3.5), on the other hand, do not lie close to the values calculated from $\sqrt{V(T; F)} \Phi^{-1}(\alpha)$ in (3.4). For instance, for $\alpha = 0.90$ and T the corresponding functional for the *HampelR* estimator with optimal tuner $b = 1.0$, we find $\sqrt{V(T; F)} \Phi^{-1}(0.90) = \sqrt{0.3947} \cdot 1.2816 = 0.8051$. The corresponding value from table 5.1 is $c_1 = 1.01$ and there is a relatively large difference between the two values. This difference could perhaps be explained by the asymptotic nature of (3.4) as the curves fitted were influenced by relatively small sample sizes (see figure 5.1).

Table 5.1 Constants for calculating quantiles when constructing confidence intervals for the scale parameter of the Rayleigh distribution.

Coverage probability		HampelR	CauchyR	MleR	MadR	TrMleR	QnR
0.90	c_0	1.01	0.93	0.81	1.05	1.08	1.79
	c_1	0.49	0.50	0.50	0.47	0.50	0.58
0.91	c_0	1.04	0.95	0.84	1.08	1.11	1.84
	c_1	0.50	0.50	0.50	0.47	0.49	0.58
0.92	c_0	1.08	0.98	0.87	1.11	1.15	1.91
	c_1	0.50	0.50	0.50	0.47	0.49	0.58
0.93	c_0	1.12	1.02	0.90	1.15	1.18	1.97
	c_1	0.50	0.50	0.50	0.47	0.49	0.58
0.94	c_0	1.16	1.06	0.92	1.20	1.25	2.05
	c_1	0.50	0.50	0.50	0.47	0.50	0.58
0.95	c_0	1.22	1.11	0.97	1.24	1.31	2.15
	c_1	0.50	0.50	0.50	0.47	0.50	0.58
0.96	c_0	1.25	1.16	1.01	1.32	1.36	2.26
	c_1	0.49	0.50	0.50	0.47	0.50	0.58
0.97	c_0	1.33	1.22	1.06	1.40	1.44	2.37
	c_1	0.49	0.50	0.49	0.48	0.50	0.58
0.98	c_0	1.41	1.32	1.14	1.49	1.56	2.58
	c_1	0.49	0.50	0.50	0.47	0.50	0.59
0.99	c_0	1.58	1.47	1.26	1.69	1.73	2.88
	c_1	0.50	0.50	0.50	0.48	0.50	0.59

6. EVALUATING AND COMPARING THE CONFIDENCE INTERVALS

To test and compare the confidence intervals based on the estimators from section 4 there are two important factors to be considered, namely the real coverage probabilities and the lengths of the confidence intervals. These two factors usually counteract with one another in the sense that short confidence intervals result in low coverage probabilities and vice versa.

A Monte Carlo study was done to test the intervals and make comparisons. We deviated from the normal practice of choosing fixed sample sizes in the following way: $N^* = 100\ 000$ samples were drawn, for $\theta = 1$ (without loss of generality) and selecting n (the sample size) at random between 10 and 100. By following this procedure we hoped to cover many possibilities in relatively few computations. From each sample the confidence interval based on the estimator under scrutiny was computed. The number of times the confidence interval covered the value of θ was counted, and this divided by N^* gave an estimate of the real coverage probability. The length of each of the N intervals was also recorded and the average of these used as an appropriate summary measure of interval length. However, it could be argued that these confidence interval lengths cannot be compared as they are based on different sample sizes. Nevertheless, since the sample sizes are spread evenly over the selected range, it appears acceptable to use the mean confidence interval length for comparative purposes. The selected nominal coverage probability was 0.95.

The simulations were based on samples arising from a pure Rayleigh distribution on the one hand as well as on certain contaminated Rayleigh distributions on the other. The robust estimators and the confidence intervals derived from them should show their quality in the contaminated case in particular. We considered two types of contamination namely outlier and inlier contamination. A contaminated Rayleigh distribution is denoted by $\alpha\%R(c\theta)$ and this notation implies a mixture of $100(1-\alpha)\%$ from a $R(\theta)$ -distribution and $100\alpha\%$ from $R(c\theta)$. The two cases we consider are $10\%R(2\theta)$ (outlier contamination) and $10\%R(\theta/10)$ (inlier contamination).

Figure 6.1 depicts the densities of these contaminated distributions. Additional choices of contamination make the task of comparing confidence intervals more unwieldy. We are of the opinion that the contamination described above is sufficiently extreme to illustrate the robustness qualities of the proposed confidence intervals.

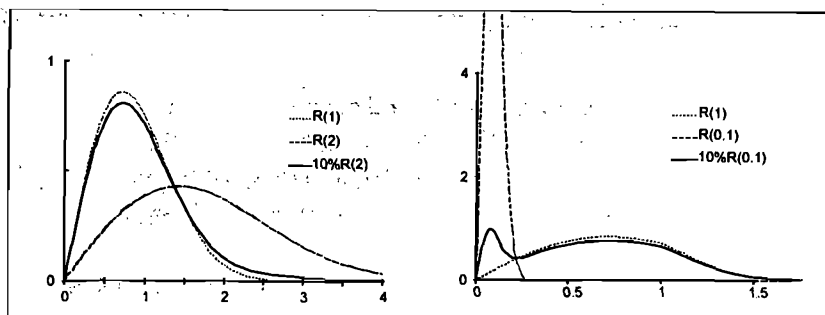


Figure 6.1. The graph on the left (solid line) shows the outlier contaminated Rayleigh density function with a tail somewhat heavier than that of the underlying distribution. The graph on the right depicts the inlier contamination leading to a bimodal distribution.

We first consider the case with an uncontaminated distribution as underlying model. In this case the nominal coverage probability and the real coverage probability are expected to be approximately equal (see table 7.1). Thus we consider it reasonable to study the interval length in isolation and comparison of the quality of estimators in this case do not present difficulties.

When the underlying distribution is a contaminated Rayleigh the problem of comparing the confidence intervals gets complicated. For a given estimator the real coverage probability now differs meaningfully from the nominal coverage probability. This has the effect that if two estimators are

to be compared at a nominal coverage probability, there may in fact be two different real coverage probabilities. This makes it meaningless to compare corresponding interval lengths in isolation because short intervals could also mean lower coverage probability and vice versa. Therefore we have to look at these two factors simultaneously and identify those confidence intervals with high coverage probability and short interval lengths.

Our strategy to solve this problem is to select a fixed estimator, compare it to each of its competitors, and identify for which of the others will it have:

- larger coverage probability AND shorter interval length,
- larger coverage probability BUT longer interval length,
- smaller coverage probability BUT shorter interval length
- smaller coverage probability AND longer interval length.

Thus to compare a given estimator T (with coverage probability P_T and interval length ℓ_T) to a competing estimator S (with coverage probability P_S and interval length ℓ_S), the interval length is plotted against the coverage probability (for the selected nominal coverage probability of 0.95) with (P_T, ℓ_T) as origin of the graph. The quadrant in which the plotted point (P_S, ℓ_S) lies, gives an indication of how T performs against S . Figure 6.2 provides the key for this interpretation.

If (P_S, ℓ_S) lies in quadrant 2, then T induces a larger coverage probability and shorter interval length than S . In this case T is clearly superior to S . If (P_S, ℓ_S) lies in quadrant 4, then T induces less coverage probability and longer interval length than S , in which case it is inferior to S . In quadrants 1 and 3 there will be no clear winner because T will have either smaller coverage probability but a shorter interval length (quadrant 1) or larger coverage probability but longer interval length (quadrant 3).

To compare all the other estimators with estimator T , the points (P_S, ℓ_S) are plotted on the graph using a unique number for identifying each estimator (see figure 7.1).

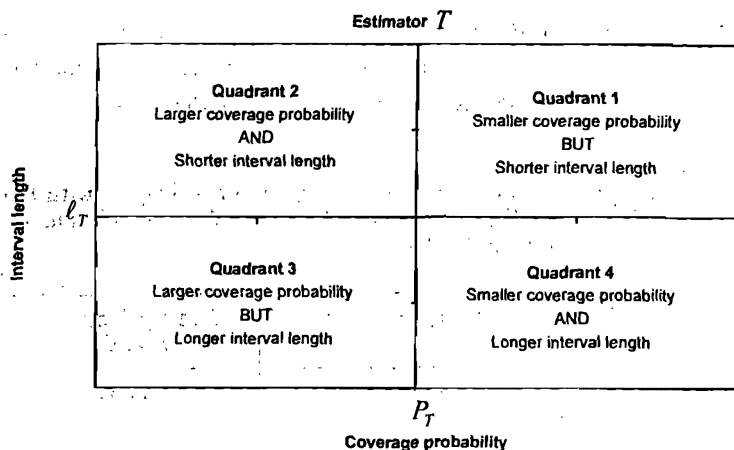


Figure 6.2. Key for interpreting the comparison graphs in figures 7.1 and 7.2. The axes cross at $(P_T; \ell_T)$.

If we follow the strategy of choosing that estimator with larger coverage probability and shorter interval length it follows that the more points appear in quadrant 2, the better the performance of the estimator. Clearly the estimator under consideration fares poorly against those plotted in quadrant 4. There is an in-between situation when points are plotted in either of quadrants 1 or 3. In that case the choice is somewhat subjective, depending on the value placed on coverage probability versus interval length.

Example: The cluster of points in the first quadrant of *MleR* in figure 7.1 is interpreted as follows:

MleR has smaller coverage probability BUT shorter interval lengths than the estimators represented by the cluster.

7. RESULTS AND CONCLUSIONS

We firstly consider the outcome of the comparison of confidence intervals in the case where the uncontaminated Rayleigh is the underlying model. Table 7.1 gives the real coverage probabilities (for nominal coverage probability of

0.95) and average interval lengths. We see that the real coverage probabilities are close to 0.95, from which we conclude that the coverage probabilities of the confidence intervals under study are approximately the same.

Table 7.1 Real coverage probabilities and average interval lengths with the uncontaminated Rayleigh as underlying distribution and nominal coverage probability of 0.95.

Estimator	Real coverage probability	Average length of interval
<i>MleR</i>	0.949	0.305
<i>CauchyR</i>	0.951	0.356
<i>HampelR</i>	0.949	0.393
<i>TrMleR</i>	0.950	0.427
<i>MadR</i>	0.949	0.453
<i>QnR</i>	0.951	0.547

The second column in table 7.1 shows, as to be expected, that the maximum likelihood estimator (*MleR*) gives confidence intervals with the shortest lengths on average. This is a direct consequence of the uncontaminated Rayleigh as underlying model where the variance of the corresponding functional for the *MleR* is equal to the Cramér-Rao lower bound. There is not much to choose between the robust estimators, *CauchyR* giving the shortest intervals.

Turning to distributions with outlier contamination, 10%R(2), we see from figure 7.1 that none of the estimators dominate any of the others (there are almost no points in quadrant 2) except for *QnR* which is dominated by all the other robust estimators. There is virtually no difference between the *HampelR*, *MadR* and *TrMleR* estimators. The *CauchyR* gives shorter interval lengths but is inferior to the others on coverage probability. To choose a best suited estimator depends on the importance of coverage probability

versus interval length. Of these we judged the former to be most important, in which case the interval of choice is any one of *HampelR*, *MadR* or *TrMleR*.

1-HampelR 2-CauchyR 3-MleR 4-MadR 5-TrMleR 6-QnR

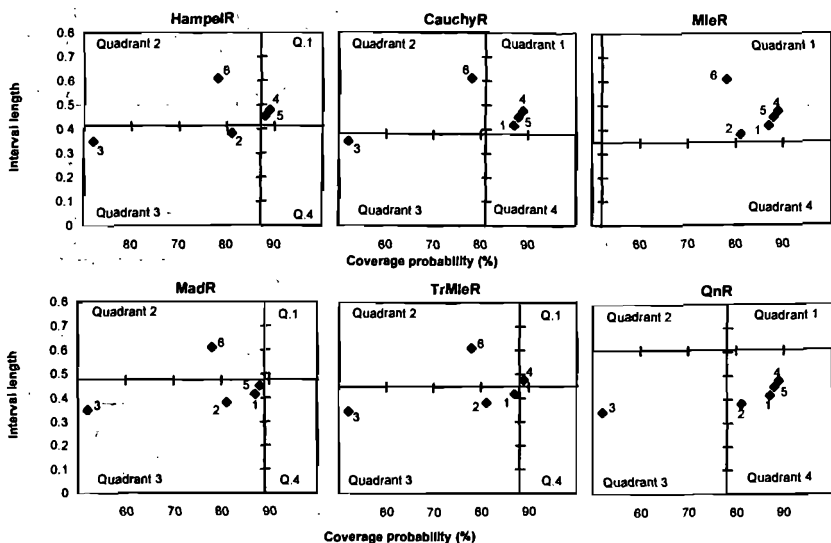


Figure 7.1. Comparing confidence intervals with outlier contamination, 10%R(2).

For inlier contamination, 10%R(0.1), we see from figure 7.2 that the *MleR* clearly dominates the other estimators. In the case of the robust estimators much the same pattern as for outlier contamination exists. Here the *MadR* and *TrMleR* in general provide larger coverage probability, but lose out on interval length.

1-HampelR 2-CauchyR 3-MleR 4-MadR 5-TrMleR 6-QnR

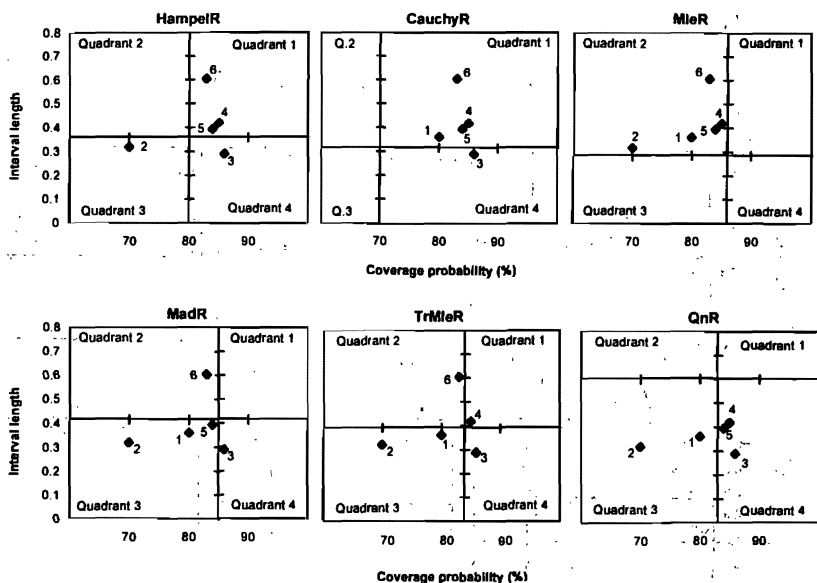


Figure 7.2. Comparing confidence intervals with inlier contamination, 10%R(0.1).

The *MleR* performed well when inlier or no contamination is present but it gives no protection against outliers and is heavily influenced by them (being a function of the sum of squares of the data). The *HampelR*, *MadR* and *TrMleR* provide appropriate protection against outliers but are outperformed by the *MleR* and *CauchyR* when no contamination occurs. As an operational rule, with no prior information about the potential for contamination, we recommend basing confidence intervals on both the *HampelR* and the *MleR*. If there are perceivable differences between the two

calculated confidence intervals, the data should be inspected before deciding which confidence interval to accept as appropriate.

Table 7.2 gives a summary of proposed estimators for the three different situations.

Table 7.2. Proposed estimators when constructing confidence intervals for the Rayleigh scale parameter

No contamination	Outlier contamination	Inlier contamination
<i>MleR</i> <i>CauchyR</i>	<i>HampelR</i> <i>MadR</i> <i>TrMleR</i>	<i>MleR</i>

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